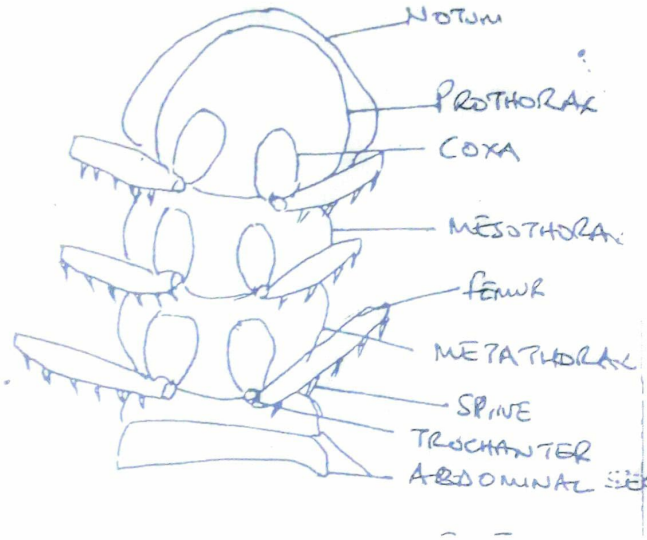
WAKISSHA
MARKING GUIDE
Uganda Advanced Certificate of Education
BIOLOGY P530/3

1. a) Features for class: - 3 thoracic segments
- 3 main body parts
- 3 pairs of limbs

(01½ marks)

b) i) DRAWING OF ANTERIOR HALF OF TRUNK OF SPECIMEN K IN VENTRAL VIEW

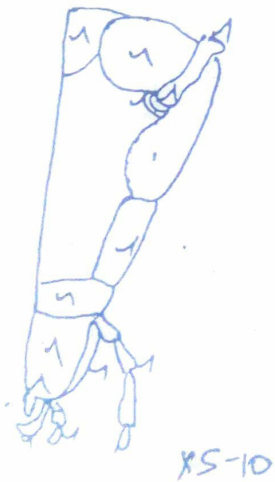


D/L = 09
M = 0½
T = 0½
N = 0½

NA - limbs including other segments
- If head included drawn below 2nd abdominal segment.

ii) Hard Notum/ Exoskeleton for protection
Jointed limb segments for flexibility during locomotion
Hard / Pointed spines for protection/locomotion on rough surfaces.

c) DRAWING OF LEFT HALF OF HEAD OF SPECIMEN K INCLUDING VISIBLE MOUTH PARTS

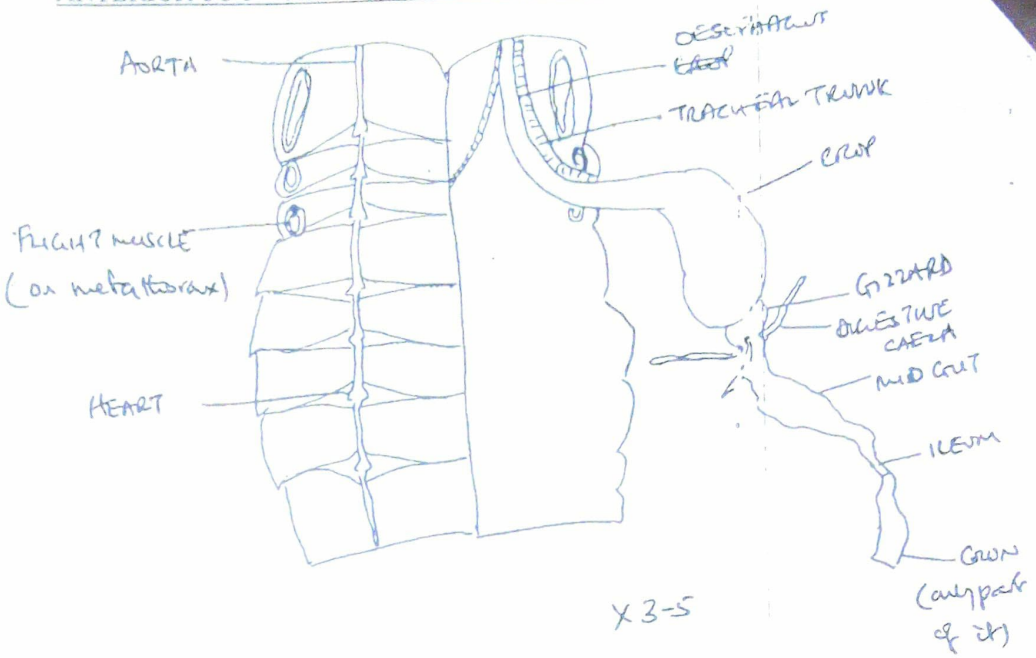


D/L = 6½
M = 0½
T = 0½
N = 0½
LFT= 01
09

NA - Right side drawn
- If drawing labeled

© WAKISSHA

d) DRAWING OF STRUCTURE FOR LOCOMOTION AND PASSAGE OF MA
ANTERIOR TO 5TH ABDOMINAL SEGMENT IN SPECIMEN K.



NA: - Prothoracic muscle labeled.
- Salivary apparatus included
- Head drawn (deny ventral cuticle)
- If Rectum, Ejaculatory duct, oviduct/vagina, mushroom shaped gland included.

D/L = 12
M = 01
T = 01
N = 01
O = 01
No Head = 01
Dorsal cuticle (left) = 01
18

2. b) sol P; since coloured drop ascended/rises; means P was dilute and less Dense; **(02 ½ marks)**

c) (i) Volume of Q used
Table 1

	1	2	3	4	5
Vol of Q in cm^3	0	4	8	12	18

@ ½ x 5 = 2½ marks

(ii) Strips curved slightly outwards towards the Epidermis;

(iii) since fleshly cortex/ tissue was turgid; but couldn't expand due to inelastic epidermis surrounding it; But on cutting the strip, the tension in the fleshy tissue was released; causing it to stretch/ expand curving over the inelastic epidermis;

(03 marks)

(iv) Table 2

Mins		10	20	30	40	50
Distance (mm)	1	55	45	35	25	0
	P	50	40	30	0	0

(05 marks)

Mark the trend i.e., P faster decrease than in1

d) i) @ ½ = 2 marks

ii) Result in Table 2

The fleshly tissue/ cortex in 1 gained water; by Osmosis; at a lower rate; since the temperature was room temp; while rate was higher at higher temp of 40^0C;

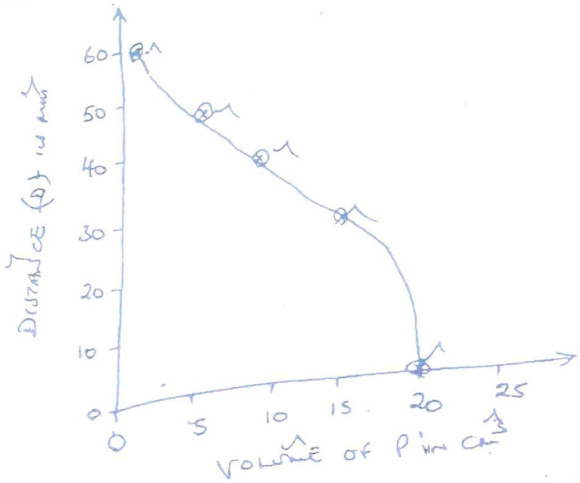
Accept OWTTE **02max**

Table 3 where (D) is < 50mm

Ext. Solutions were hypotonic to (cell sap) in fleshy tissue/ cortex; causing it to gain water; by osmosis; and expand; But the inelastic Epidermis limited the expansion; thus the observed curvature determining value of (D) **(05 marks)**

iii) 3/4; where curvature of the strip(s) is same as when it was immediately cut from R **(02 marks)**

iv) GRAPH SHOWING RELATIONSHIP BETWEEN (D) IN mm AND VOLUME OF SOLUTION P USED IN Cm^3.



Title = 01
Plotting = 02½
Curve = 01
NT = 0 ½
Scale = 01@ ½
Axes = 02@1
Total = 08

3. a) i) Description of structure of inflorescence of X

NB: stop marking where non-structural aspect given e.g. or floral parts are described. Colour

X has numerous florets; of 2 types; ray florets at outer part; tubular florets inner part; florets at the expanded apex of the pesude; and closely packed; with tubular florets at higher at higher level than ray florets; the florets are surrounded by involucre of bracts. The florets are arranged in circular pattern **(04 marks)**

© WAKISSHA

ii) Visible parts of ray florets:

Corolla- free at apex/ fused base; veined them; free apex is broad

Calyx- Many; long; thin; and hairy

Ovary-long; thin; interior ;with sutures/ ;lines of weakness

(04 marks)

b) i) and ii) 2 structural uniqueness and role.

X - forked stigma; - Large S.A for pollens;

Inferior ovary;

2 types of florets;

Long thin calyx;

Any 1st 2

Y- Staminoids;

Long/curved ovary;

Perianth; - Broad to protect inner parts;

Any 1st 2

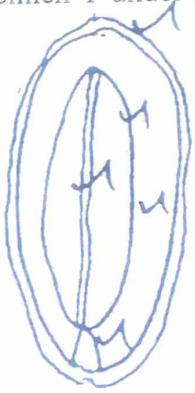
Z- Stamens attached onto Corolla; - stamens Protected;

Nectar guides present; - Eases pollination;

Any 1st 2

c) Drawing of T/S of ovary in specimen Y under low Power magnification

- Epicarp
- Mesocarp
- 2cotyledons
- gap between cotyledons
- Funicle.



X 20 – 50

D = 02½

M = 0½

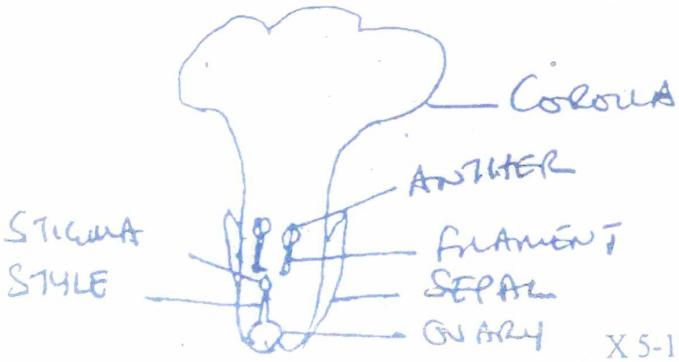
T = 0½

N = 0½

Oval shape = 0½

04½

d) DRAWING SHOWING L/S ONE HALF OF SPECIMEN Z FROM INNER VIEW



X 5-10

D/L = 07

M = 0½

T = 0½

N = 0½

08½

END

© WAKISSHA